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A Comparative Study: Various Modeling (Empirical, Integral, 3D CFD) Approach for Accurate Consequence Analysis and Risk Management of Accidental Release of Ammonia

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Abstract

Ammonia and hydrogen are increasingly recognized as key enablers of a net-zero carbon future, as both emit no carbon during consumption. With the global shift toward a hydrogen economy gaining momentum, ammonia stands out as a practical solution for hydrogen storage and transportation due to its favorable thermodynamic properties. However, ammonia's highly toxic and explosive nature brings significant safety challenges, especially in industrial environments. Ensuring its safe storage, handling, and transport requires thorough hazard identification and detailed risk assessment.

Across the world, past incidents involving ammonia leaks, toxic dispersion, and explosions have led to multiple fatalities and severe injuries. With its expanding use in energy and industry sectors, the risk of such accidents is growing making it more important than ever to enhance our understanding of the consequences and associated risks.

Integrating consequence analysis at the early stages of project development such as the concept or design phase can significantly improve decision-making. This proactive approach enables the implementation of effective safety measures, loss prevention systems, and emergency response plans from the outset, reducing both the likelihood and impact of potential incidents.

Several tools are available to model accidental releases and predict their consequences, including Gaussian dispersion models (1D), empirical formulas (2D), and Computational Fluid Dynamics (CFD) simulations in 3D. While physical experiments provide accurate results, they are costly and involve inherent risks. Likewise, 2D models struggle to replicate complex real-world conditions such as plant congestion, terrain variations, and wind flow. In contrast, 3D-CFD models can capture these complexities with high accuracy, offering realistic simulations of gas behavior under various release scenarios.

This paper presents a hypothetical case study of accidental ammonia release, focusing on gas dispersion and explosion potential using advanced 3D-CFD simulations. The study also compares outcomes from 2D and 3D models/ tools to demonstrate the added value of high-fidelity modeling in predicting real-world consequences. The results support a more informed, inherently safer design approach and optimized risk reduction measures while ensuring compliance with 100% HSE goals and minimizing unnecessary capital expenditure.

Keywords: Ammonia dispersion, Hydrogen economy, 3D-CFD modeling, Risk assessment, Consequence analysis, 100% HSE, Toxic gas release, Explosion modeling, inherently safer design, Emergency response, Process safety, Industrial hazard, CAPEX optimization

Introduction

The chemical and process industries are fundamental to supporting the increasing global demand for energy, food, and essential commodities. With the world population rising and industrialization accelerating, chemical plants have grown in both scale and complexity, which inherently increases the risk of hazardous accidents involving toxic, flammable, or explosive substances. Historical industrial disasters, such as the 1984 Bhopal gas tragedy—which claimed over 2,500 lives and caused lasting health impacts for hundreds of thousands—and the 2020 styrene gas leak in Vizag, India, which resulted in multiple deaths and thousands of injuries, underscore 1,2 the critical need for stringent safety management in hazardous industries^{1,2}

Among the many chemicals used worldwide, ammonia stands out for its widespread industrial applications, including fertilizers, refrigeration, and emerging roles in energy systems. Despite its utility, ammonia is a highly toxic gas with serious health risks even at low concentrations. It is colorless with a pungent odor and can become heavier than air in moist conditions, increasing the hazard radius upon release. Exposure limits set by occupational safety organizations, such as NIOSH, highlight the narrow margin between safe and dangerous ammonia concentrations, with levels above 300 ppm causing severe respiratory and ocular damage and levels above 2,500 ppm posing a risk of fatality³.

Ammonia's hazardous nature has led to numerous catastrophic accidents globally, including the 1992 incident in Dakar caused 129 deaths and over 1,100 injuries due to a storage failure. In China (2013), two separate leaks led to a combined 127 deaths. In India, leaks at facilities such as IFFCO (2020) and Fujairah's Ice Factory (2018) highlight continuing risks associated with poor maintenance and equipment failures⁴⁻⁶. These events highlight the persistent risks associated with ammonia handling and the urgent need to advance safety measures.

In the context of global efforts to combat climate change, ammonia has gained increasing attention as a clean energy carrier. It plays a critical role in hydrogen storage and transport, which is vital to achieving the Paris Agreement's 2050 carbon neutrality goals. The International Maritime Organization (IMO) recognizes ammonia as a promising marine fuel candidate to reduce greenhouse gas (GHG) emissions from shipping—one of the hardest sectors to decarbonize—with ambitious targets of reducing GHG emissions by at least 20% by 2030 and 70% by 2040 compared to 2008 levels⁷⁻⁹.

Ammonia offers several advantages for marine propulsion beyond zero carbon emissions during combustion. It is compatible with various engine technologies, including combustion engines and fuel cells, and benefits from established production and supply chains primarily centered on fertilizer manufacturing. However, current large-scale ammonia production is mostly fossil fuel-based, resulting in substantial GHG emissions. The development of "green ammonia," produced from renewable energy sources, remains costly and technologically challenging.⁷⁻⁹

Despite its potential, ammonia's toxicity and corrosiveness pose significant safety challenges, particularly for marine applications where accidental releases can have severe consequences. Ammonia-powered engines also risk emitting nitrogen oxides (NOx) and nitrous oxide (N₂O)—the latter being a potent greenhouse gas—potentially offsetting some environmental benefits depending on combustion control^{7,10,11}.

Safety regulations for ammonia use in marine environments are still in early stages. While industries like petrochemicals and fertilizers have extensive experience handling ammonia safely on land, the use of ammonia as ship fuel introduces new risks that require dedicated risk assessment and mitigation strategies. Studies stress the urgent need for international regulatory frameworks tailored to ammonia propulsion, given its high toxicity and unique operational hazards¹².

Understanding ammonia release behavior is crucial. Large-scale ammonia leaks can create toxic clouds capable of traveling long distances and causing catastrophic health effects. Experimental studies, such as those by Yarandi et al. (2021)¹³, showed toxic impacts extending up to 920 meters from industrial ammonia release sources, demonstrating the severity of potential marine fuel leak consequences¹³. Other investigations into ammonia dispersion in food processing and fertilizer facilities reveal lethal concentration zones within tens to hundreds of meters, emphasizing the importance of accurate modeling and emergency preparedness^{4,5,14}.

Accidental ammonia releases in marine settings—especially involving collisions or hull breaches—can result in rapid, complex dispersion patterns affected by water dissolution, atmospheric conditions, and vessel design. While there is extensive research on onshore ammonia leaks, studies addressing offshore scenarios remain limited.

Traditionally, experimental studies such as the Thorney Island (UK, 1984) and China Lake (USA, 1987) tests provided key data for validating dispersion models. However, such experiments are costly, environmentally risky, and difficult to repeat under controlled variables like wind speed, direction, and terrain. As a result, most recent research has shifted toward modeling approaches⁵.

Given the high costs and risks of large-scale experimental testing, Computational Fluid Dynamics (CFD) has become an indispensable tool for simulating ammonia dispersion and explosion consequences. Unlike simpler empirical or 2D models, 3D-CFD simulations capture complex turbulent flows, plant congestion, terrain effects, and meteorological variability with high fidelity. These simulations provide detailed insights crucial for designing inherently safer facilities, optimizing mitigation measures, and informing emergency response planning, thereby reducing capital expenditures while maintaining robust health, safety, and environmental standards.

This paper presents a comprehensive review of recent advances in CFD applications for ammonia consequence analysis, with a focus on industrial and marine case studies. A hypothetical ammonia release scenario is modeled using advanced 3D-CFD tools, comparing results against 2D models to demonstrate enhanced accuracy and risk assessment capability. The findings contribute to the evolving knowledge base required for safe ammonia adoption in emerging low-carbon energy and maritime fuel sectors.

Literature Review: Urgency for CFD-Based Consequence Analysis in Ammonia Risk Management

Several researchers have employed various modeling approaches—ranging from empirical and integral models to semi-empirical software tools like PHAST and ALOHA—to predict dispersion behavior. These methods, while useful for preliminary estimates, often fall short in capturing complex fluid dynamics, 3D geometrical influences, and real-time atmospheric variations. This limitation becomes particularly significant for ammonia, whose behavior is sensitive to humidity, temperature, and terrain characteristics.

Ng et al. (2023) utilized PHAST to model ammonia dispersion during bunkering operations in Singapore. Their work demonstrated how ammonia dispersion over seawater can travel farther than over land due to the exothermic dissolution process, which enhances vaporization. The study emphasized the influence of operational parameters such as release height, direction, and storage condition on toxicity footprints. While PHAST offered valuable scenario analysis, the model operated within simplified assumptions—highlighting its limitations for capturing site-specific effects in dynamic, obstacle-rich environments¹⁵.

To overcome these limitations, Kim and (2024) adopted high-resolution CFD simulations using PyroSim and Star-CCM+ to investigate ammonia venting from storage tanks aboard a ship. Their results confirmed that the vent height and ship geometry significantly influenced dispersion patterns. Unlike empirical models, CFD captured spatial variability and the influence of ship structure on gas movement—demonstrating its superiority in real-world safety design¹².

Similarly, Namboothiri and Soman (2018) combined CFD-based pool evaporation modeling with ALOHA dispersion and probit analysis to assess population exposure risks. Their multi-tool approach allowed for a broader consequence picture, but the reliance on ALOHA for dispersion still introduced simplifications, particularly in terrain and obstacle handling¹⁶.

In comparison-focused studies, Guddal (2025) directly validated FLACS-CFD and PHAST against INERIS ammonia release experiments. The study found that while both tools performed adequately in open terrain, **CFD significantly outperformed PHAST in obstacle-dense environments due to its 3D spatial resolution and ability to simulate complex interactions between gas, structures, and wind flow**. This finding critically supports the need to use CFD tools for accurate consequence analysis in realistic maritime and industrial settings¹⁷.

Further, Bae et al. (2025) and Duong et al. (2025) applied FLACS to study explosion characteristics in confined compartments on ammonia-fueled ships. Their results identified sensitive parameters like ignition location, leak direction, and equipment layout—all of which would be impossible to assess using traditional integral models or 2D tools^{8,9}.

Tan et al. (2017), Gangopadhyay et al. (2008), and Roy et al. (2011) have relied on statistical, empirical, and fuzzy logic-based methods, which, while useful for preliminary hazard assessments, lacked the granularity needed to model toxic cloud evolution and real-time dynamics in accident scenarios. These limitations further demonstrate the insufficiency of legacy tools when high-risk fuels like ammonia are involved^{3,4,18}.

Several CFD-specific studies—including those by Rebecca et al. (2018), Govind et al. (2022), and Sutton et al. (1986)—emphasized how turbulent flow, leak orientation, ventilation obstructions, and ignition location dramatically alter risk outcomes. These dynamics cannot be replicated in tools like PHAST, which apply fixed source strength and dispersion assumptions^{2,19,20}.

Despite this growing body of CFD research, **very few comparative studies have systematically analyzed the discrepancies between empirical tools, semi-empirical models, and full CFD simulations for ammonia release scenarios. This constitutes a major gap in the literature** gap that this paper seeks to address by conducting a direct comparative assessment of multiple modeling approaches (empirical, integral, PHAST, and 3D CFD) under identical release conditions.

This paper demonstrates that traditional tools consistently underpredict or oversimplify dispersion extents, toxicity zones, and the influence of real-world geometry. For high-fidelity consequence analysis—especially in marine terminals, industrial clusters, or enclosed vessel spaces—**CFD emerges as the only viable modeling approach capable of delivering reliable and site-specific safety insights**.

In conclusion, while empirical and integral models offer speed and ease of use, **they fall short in accuracy, adaptability, and scenario realism**. CFD tools, particularly Ansys, GEXCON FLACS, PyroSim, and Star-CCM+, provide the precision, flexibility, and validation fidelity needed for today's risk environment. With ammonia poised for widespread use as a low-carbon fuel, adopting CFD-based risk modeling is not just recommended—it is imperative.

Methodology and Numerical approach

This study presents a hypothetical case of accidental ammonia release from a storage system, with a focus on modeling the dispersion behavior using advanced 3D Computational Fluid Dynamics (CFD) techniques. The simulation explores the influence of key environmental and design factors, including terrain topology,

low wind conditions, crosswinds, and structural obstacles, and contrasts the results with those obtained from a traditional 2D modeling approach.

Previous research by Efthimiou et al. (2017) and Sharma et al. (2020) categorizes consequence modeling techniques into three primary types: (a) Gaussian models, (b) Integral or Lumped Element Models (LEM), and (c) Computational Fluid Dynamics (CFD) models^{21,22}. Among these, Gaussian models are the most basic and assume flat terrain and stable meteorological conditions (e.g., constant wind speed and direction), which limits their applicability in real-world scenarios^{23,24}.

Integral models (or LEMs) use simplified one-dimensional equations based on conservation laws for mass, momentum, and energy²¹. They are computationally efficient and user-friendly, making them suitable for dispersion studies in unobstructed environments. However, their accuracy significantly diminishes in the presence of complex geometries or obstacles²⁵.

In contrast, CFD provides a robust, high-fidelity approach to simulate hazardous gas dispersion under realistic conditions. Unlike the one-dimensional nature of Gaussian and Integral models, CFD offers a full three-dimensional analysis, capable of capturing the interactions between the gas cloud and surrounding terrain, structures, and atmospheric turbulence. While both Integral and CFD models may yield comparable results in open areas, CFD excels in complex industrial environments where geometry significantly influences dispersion behavior²¹.

Major Regular tools like ALOHA, Effect, FDS, PHAST, Riskcurve, and SAFETI rely on Gaussian and Integral models, making them suitable primarily for idealized, open-space conditions²⁶. However, they may fail to accurately represent the turbulent behavior of toxic gas releases in congested environments.

Anhydrous ammonia, typically stored under high pressure and low temperatures, presents a unique challenge when released. Its dispersion involves rapid vaporization, air mixing, and potential toxic exposure, forming a highly complex flow pattern that general modeling tools cannot adequately capture²⁷.

Gexcon's FLACS CFD platform, developed over four decades and validated through numerous experiments and international standards, is purpose-built for such high-risk scenarios. As demonstrated in Hansen et al. (2010), where 33 experimental cases were simulated and validated using the Model Evaluation Protocol (MEP), FLACS has shown exceptional accuracy in predicting gas dispersion behavior²⁸.

Numerical Codes and Machine

In the present study, **Gexcon's FLACS 3D CFD software (Version 22.2)** has been employed for detailed consequence analysis related to ammonia dispersion and ventilation scenarios. All simulations were executed on a high-speed computing system to ensure efficiency and accuracy in processing complex geometries and atmospheric conditions. For comparison purposes, **Gexcon EFFECT** (which incorporates Gaussian-based models for passive dispersion) and **DNV's Phast** (based on integral and empirical dispersion models, but not purely Gaussian) were used to perform simplified consequence assessments. The empirical and integral model results provided a baseline for comparative evaluation, especially in open environments with minimal obstruction.

While integral and empirical models are included to provide comparative insights, the **primary focus of this paper is on the FLACS CFD modeling approach**, given its superior capability to accurately simulate complex dispersion behavior, including the impact of obstacles, terrain variations, and atmospheric dynamics. For this study, **Gexcon's own licensed tools—FLACS CFD and EFFECTS—were utilized**, whereas the **DNV Phast software** was accessed by the **licensed facilities of the research group at UPL University**, who also contributed to model execution, data interpretation, and comparative analysis. As a result, the detailed methodology section is dedicated to the FLACS CFD tool, while the outputs from EFFECTS and Phast are presented in the results section for benchmarking and validation purposes.

FLACS CFD Computational Methods

FLACS CFD code works in 3 steps: Preprocessing (Geometry, Source Data calculations and Simulation), Processing (Numerical Iterations), and Postprocessing (Result visualization)

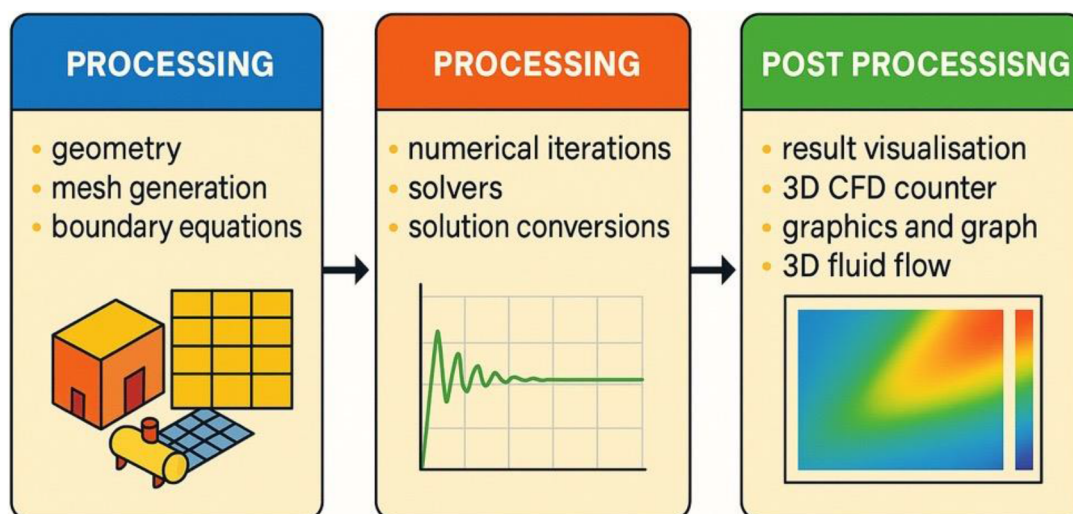


Figure 1—FLACS CFD process flow diagram

Source Term Data Calculation for Toxic Ammonia Gas Release

In this study, a hypothetical ammonia leak scenario was modeled using different modeling tools (PHAST, EFFECTS, and FLACS CFD) to understand the impact of varying storage pressures and temperatures on release rates. The release parameters were defined to represent a credible small-bore leak from a pressurized ammonia storage system. A circular orifice of 10 mm diameter was selected to represent the leak point, with a discharge coefficient of 0.96, assuming a rounded edge orifice. For terrain impact, two ground roughness values were considered 0.03 m (low terrain) and 0.10 m (medium terrain). The simulation utilized wind and nozzle-based boundary conditions to control fluid flow behavior at the release point. The detailed assumptions are given in the below [Table 1](#)

Table 1—Modeling Assumptions Adopted in EFFECTS, PHAST, and FLACS

Parameter	PHAST (Integral)	EFFECTS (Empirical)	FLACS CFD (3D CFD)
Release Type	Pressurized leak from ammonia storage		
Source Term	Calculated internally in EFFECTS; simplified jet/plume model.	Source Term calculated internally using PHAST dispersion module.	Source Term calculated internally using FLACS and additionally imported from PHAST/EFFECTS and simulated within CFD solver.
Storage Conditions	3 bara @ -1.7°C , 5 bara @ 9.43°C , 8 bara @ 21.68°C		
Ambient Conditions	25 $^{\circ}\text{C}$, 1 bara	25 $^{\circ}\text{C}$, 1 bara	25 $^{\circ}\text{C}$, 1 bara
Atmospheric Stability	Pasquill Class 2F (moderately stable)		
Wind Stabilization Time	NA	NA	100 s
Wind Directions	90 $^{\circ}$ (upwind/downwind)		270 $^{\circ}$ (crosswind), 90 $^{\circ}$ (upwind/downwind)
Terrain Roughness	Not explicitly modeled		0.03 m (low), 0.10 m (medium)
Obstacles / Terrain Effects	NA		Explicitly modeled (buildings, terrain, crosswind interactions)
Boundary Conditions	Simplified integral solution	Simplified empirical solution	Wind and nozzle boundary conditions applied
Hight of leak	1 m from ground		At actual hight of the leak source as per 3D model

To compare the performance of various consequence modeling tools, three distinct cases were analyzed with fixed volume (5 m³) and varying temperature and pressure conditions. The release rates (in kg/s) predicted by PHAST, EFFECTS, and FLACS CFD are provided below:

Table 2—Release rate comparison (EFFECTS, PHAST, and FLACS) under varying pressure and temperature conditions.

Case	Volume (m ³)	Temperature ($^{\circ}\text{C}$)	Pressure (bara)	Release Rate (kg/s)		
				EFFECTS	PHAST	FLACS
1	5	-1.7	3	0.03586	0.0478	0.055
2	5	9.43	5	0.0556	0.07202	0.080
3	5	21.68	8	0.0846	0.10777	0.120

Preprocessing- Geometrical modeling and Leak location Development

In the present study, three hypothetical geometry configurations were considered to evaluate the extent of gas dispersion using a **3D- CFD approach**. The objective was to investigate the influence of various

parameters, including obstacles, crosswinds, terrain effects, and low wind conditions, on fluid flow paths. The geometries were defined as follows:

- **Geometry 1:** No obstacles in the fluid flow path.
- **Geometry 2:** Obstacles present within the fluid flow path.
- **Geometry 3:** Natural terrain presents in the fluid flow path.

These configurations are illustrated in the geometry layouts shown below. The geometries were developed in the **FLACS CASD**, incorporating various cylindrical and box-shaped elements to represent the ammonia facility, surrounding process areas, and sensitive receptors. The same geometry layouts were also used in the **integral and empirical modeling tools** to maintain consistency, enabling a direct comparison of results between the **3D CFD, integral, and empirical approaches**.

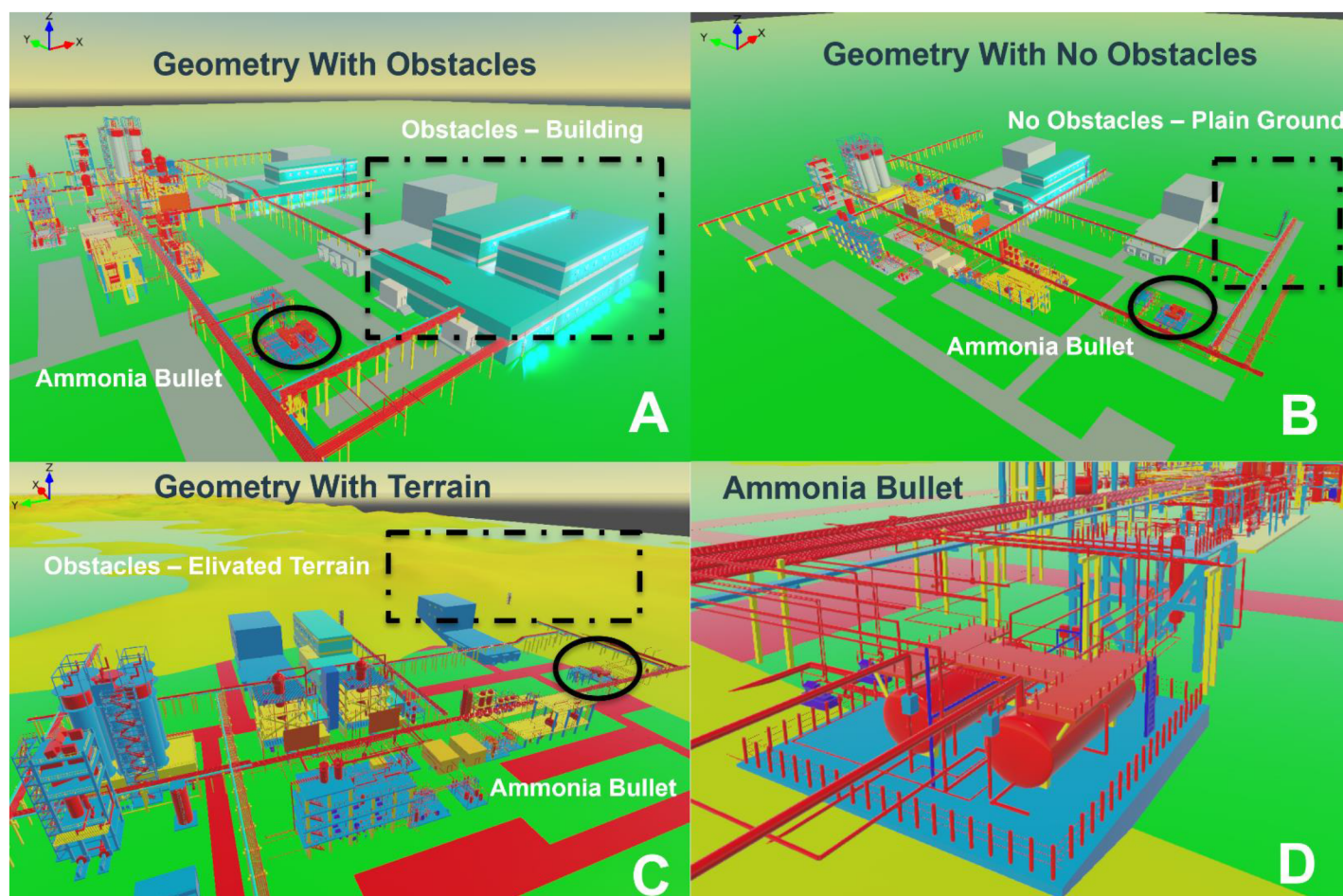


Figure 2—(A) Geometry with Obstacles - Building, (B) Geometry with No Obstacles - Plain Ground, (C) Geometry with Obstacles - Building, (D) Ammonia Storage bullets

Preprocessing- Simulation setup (initial and boundary conditions)

The boundary conditions applied in this study included WIND (inflow and parallel boundaries) and NOZZLE (exit boundary) configurations, where the outlet boundary is considered more robust and stable compared to the EULER approach [18]. Two wind directions were analyzed: 270° (downwind from the ammonia bullets towards the obstacles) and 90° (crosswind towards the ammonia bullets). The wind direction was aligned parallel to the default cartesian coordinate system; hence, XLOW, YLOW, XHIGH, and ZHIGH were defined as WIND boundaries, while YHI and ZLO were defined as NOZZLE boundaries.

The relative turbulence intensity (RTI) was set at 0.1, with the turbulence length scale (TLS) considered as 10% of the RTI. A wind stabilization time of 100 seconds was applied. All these boundary conditions and parameter values were selected in accordance with FLACS standard modeling guidelines to ensure realistic and consistent simulation results.

CFD Computational Domain

The **structured fine uniform grid** selected for the simulation was **1 m**, ensuring a balance between computational accuracy and efficiency. **Grid stretching** of up to **8 m** was applied in the far-field region to optimize computational resources. The computational setup comprised a **core domain** of **249 m × 190 m × 12 m** and a **stretched domain** of **400 m × 366 m × 70 m** in the X, Y, and Z directions, respectively. Additional **grid refinement** was applied to the region located **downwind/upwind from the release source** to improve resolution in the areas most affected by the dispersion. In **FLACS CFD**, the **Courant-Friedrich-Lewy (CFL) number** is used to link the simulation time step to the control volume (CV) dimensions through the signal propagation velocity, considering both the **speed of sound** and the **fluid flow velocity**. These are defined as **CFLC** (acoustic-based) and **CFLV** (velocity-based) parameters.

For the present **ammonia dispersion study**, values of **CFLC = 10** and **CFLV = 1** were selected in accordance with the FLACS recommended guidelines for dispersion and ventilation analyses.

Governing Equations

In the present study, **Favre-averaged transport equations** for **momentum**, **mass**, **enthalpy (h)**, **turbulent kinetic energy (k)**, **rate of dissipation of turbulent kinetic energy (ε)**, **fuel mass fraction (Y_F)**, and **mixture fraction (ξ)** are solved to capture the transient gas dispersion behavior. The **standard k-ε turbulence model**²⁹ is employed, which is based on the **Reynolds-averaged Navier-Stokes (RANS)** formulation. In this approach, *k* represents the transport equation for turbulent kinetic energy, while *ε* governs the dissipation rate of turbulent kinetic energy. This model is widely used in industrial CFD applications due to its robustness, relatively low computational cost, and ability to provide reasonably accurate predictions for atmospheric dispersion scenarios involving moderate to high Reynolds number flows. For the present ammonia release simulations, the model constants (*C_μ*, *C1_ε*, *C2_ε*, *σ_k*, *σ_ε*) were kept at their standard values as recommended in FLACS best practice guidelines, ensuring consistency and comparability with previous industrial-scale validation studies.

FLACS Validation for Ammonia Dispersion

Gexcon has developed an automated validation framework for FLACS-CFD, comprising more than 60 validation series and over 2,700 simulations, ranging from basic verification cases to large-scale gas dispersion and explosion scenarios. FLACS is a well-established and internationally recognized CFD tool for gas dispersion and explosion modeling, benefiting from more than four decades of development, continuous validation, and certification by authorities. The work of Hansen et al. demonstrated that FLACS successfully reproduced the outcomes of 33 dispersion experiments contained in the Model Evaluation Protocol (MEP) database, meeting or exceeding the recommended statistical performance criteria³⁰.

FLACS-CFD has been validated for ammonia dispersion through several large-scale experimental campaigns and real-world applications. The most significant validations come from international field trials such as the Desert Tortoise experiments³¹ and the FLADIS trials³², where large pressurized and liquefied ammonia releases were carried out under controlled conditions. These datasets provided benchmark measurements of near-field jet/aerosol dynamics, dense-gas slumping, and far-field dispersion. FLACS simulations of these trials showed good agreement with observed concentration profiles and dispersion trends, confirming its ability to reproduce both two-phase jet source terms and atmospheric dispersion regimes.

More recently, FLACS has also been applied to the Red Squirrel ammonia field tests at DNV Spadeadam³³, which investigated refrigerated and pressurized ammonia releases on land and water. These trials confirmed FLACS' ability to reproduce dispersion footprints and highlighted sensitivities to meteorological conditions, release orientation, and rainout behavior.

FLACS has further been used in practical consequence studies, including ammonia dispersion in industrial plants³⁴, refrigerated storage spaces [6], and marine fuel preparation rooms. In these applications, validation against large-scale trials underpins confidence in its predictive performance.

Specific to ammonia, Tan et al.^{35,36} FLACS predictions against small-scale wind tunnel experiments and field data, showing that ammonia dispersion is highly sensitive to release rate, wind speed, and release height. These studies observed that ammonia plumes tend to rise near the source due to buoyancy and that obstacles can significantly increase local concentrations on the windward side, emphasizing the importance of geometry in risk assessments. Duong et al.³⁷ further compared ammonia with LNG dispersion, finding that ammonia has a longer dispersion distance and persistence, necessitating larger safety zones.

Comparative analyses within the Jack Rabbit III model inter-comparison study³⁸ demonstrated that FLACS predictions align well with experimental data and are consistent with other advanced CFD models.

Overall, the validation evidence demonstrates that FLACS is a robust CFD tool for ammonia dispersion analysis, capable of capturing critical physical effects such as two-phase jet dynamics, aerosol rainout, obstacle interaction, and atmospheric variation —making it suitable for consequence analysis and quantitative risk assessment (QRA) of ammonia handling, storage, and bunkering operations. Together, these studies confirm that FLACS-CFD provides a validated, reliable, and internationally accepted platform for simulating ammonia dispersion in both open and confined environments. Its demonstrated capability to reproduce real-world ammonia release scenarios underpins its use in consequence analysis and QRA for industrial safety and emerging applications such as ammonia bunkering and fuel handling.

Simulation matrix and scenario combinations

For the current analysis, **multiple scenarios and combinations** were performed with varying inlet conditions, involving releases from **three different process conditions**. The release rates were calculated using **three different tools** with **three corresponding geometrical models**. In the **EFFECTS** and **PHAST** tools, three different parameter combinations (as listed in the respective table) were applied to estimate the release rate and evaluate the influence of these parameters on gas dispersion in **1D and 2D layouts**. In total, six cases were carried out three in PHAST and three in EFFECTS to assess the performance of **integral** and **empirical** modeling approaches.

Similarly, the same three scenarios were simulated in **FLACS Leak Wizard** to estimate the source term data for each case. The source term data from the six EFFECTS/PHAST cases were also directly applied in FLACS to ensure consistency of release conditions, resulting in **nine CFD simulations** (three from EFFECTS, three from PHAST, and three directly from FLACS Leak Wizard).

In addition, these cases were further considered in **3D geometry configurations** with and without obstacles and terrain. Overall, a total of **20 CFD simulation cases** were performed, covering all combinations of modeling approaches, geometrical complexities, and scenario variations.

Results and Discussion

As discussed above, to compare **2D (integral and empirical) results** against **3D CFD models** and demonstrate enhanced accuracy and risk assessment capabilities, multiple combinations were prepared using both tool types. The results from each approach were compared directly. The comparative assessment was performed through the following cases:

- **Case 1:** Integral and empirical (**EFFECTS**) vs. 3D CFD (**FLACS**) - Source Term Model (STM) derived from EFFECTS.

- **Case 2:** Integral and empirical (**PHAST**) vs. 3D CFD (**FLACS**) - Source Term Model (STM) derived from PHAST.
- **Case 3:** Source Term Model (STM) from **3D CFD FLACS** with terrain, and crosswind effects.

Case 1: Comparison of Integral/Empirical (EFFECTS) vs. 3D CFD (FLACS)

In this case, the Source Term Model (STM) was first derived using the EFFECTS software. The STM was then applied in both EFFECTS and FLACS simulations for direct comparison. In EFFECTS, a single dispersion case was performed, and the results were mapped onto the site layout for Ammonia IDLH value (Fig. 3). As EFFECTS is a 1D/2D tool, it does not account for full 3D geometry or obstacle effects. The same source term parameters were then used in FLACS to simulate dispersion over a 3D geometrical model, both with and without obstacles, enabling a comparative assessment.

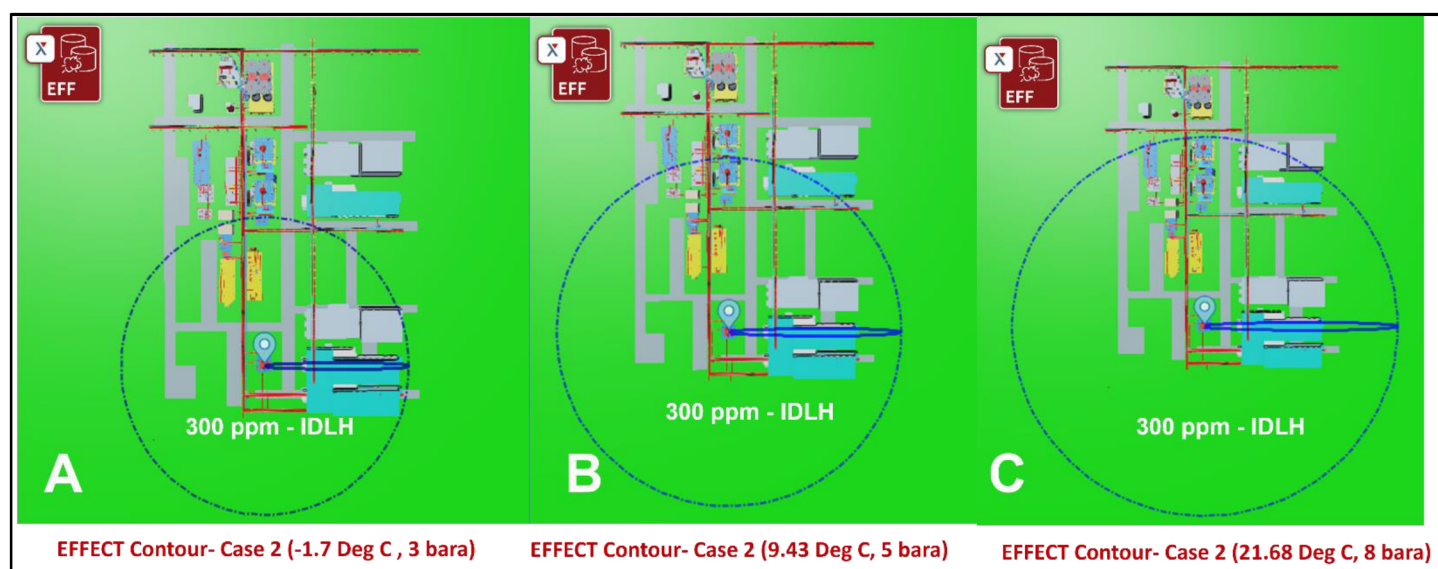


Figure 3—Ammonia 100 ppm IDLH gas dispersion counter from EFFECTS (Integral/Empirical) tool on Layout shows substantially longer dispersion distances because it does not consider physical obstacles that restrict gas movement

The comparison of EFFECTS (2D) with FLACS (3D) shows a significant difference in predicted downwind effect distances given in the Table 3–Table 5. EFFECTS predicts substantially longer dispersion distances because it does not consider 3D geometrical features or obstacles that restrict gas movement and can be seen in the Fig. 3 and Table 3.

Table 3—EFFECTS (Integral/Empirical) Dispersion Distance Results - 2D Layout

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
1	5	-1.7	3	0.03586	222	10
2	5	9.43	5	0.0556	283	12
3	5	21.68	8	0.0846	343	14

Table 4—FLACS CFD - No Obstacles Condition (Using STM from EFFECTS) Dispersion Distance Results

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
100000	5	-1.7	3	0.03586	40	9.5
200000	5	9.43	5	0.0556	57	15
300000	5	21.68	8	0.0846	60	16

Table 5—FLACS CFD - With Obstacles Condition (Using STM from EFFECTS) Dispersion Distance Results

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
111111	5	-1.7	3	0.03586	24	17.5
222222	5	9.43	5	0.0556	25	30.25
333333	5	21.68	8	0.0846	57	34

When the same STM is applied in FLACS without obstacles, the predicted downwind distances are considerably shorter and can be seen in the [Fig. 4](#) and [Table 4](#), indicating the impact of full 3D atmospheric and turbulence modeling.

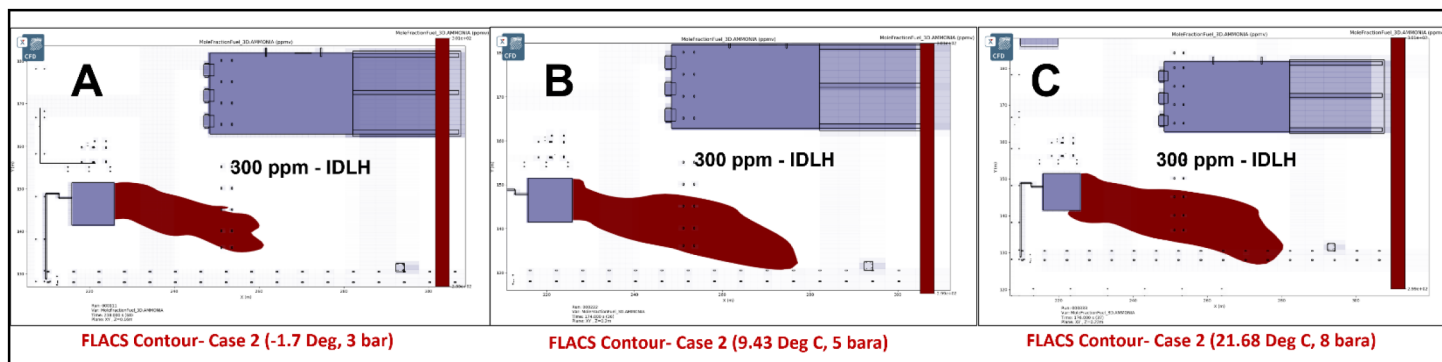


Figure 4—Ammonia 100 ppm IDLH gas dispersion counter from FLACS CFD 3D counter plot for no obstacle condition shows downwind distances are considerably shorter indicating the impact of full 3D atmospheric and turbulence modeling.

In the FLACS simulations with obstacles, the downwind distance is further reduced due to physical obstruction of the gas plume and can be seen in the [Fig. 5](#) and [Table 5](#). However, orthogonal (crosswind) spread increases because obstacles cause turbulence, eddy formation, and gas accumulation near surfaces—effects that cannot be captured by a 2D model.

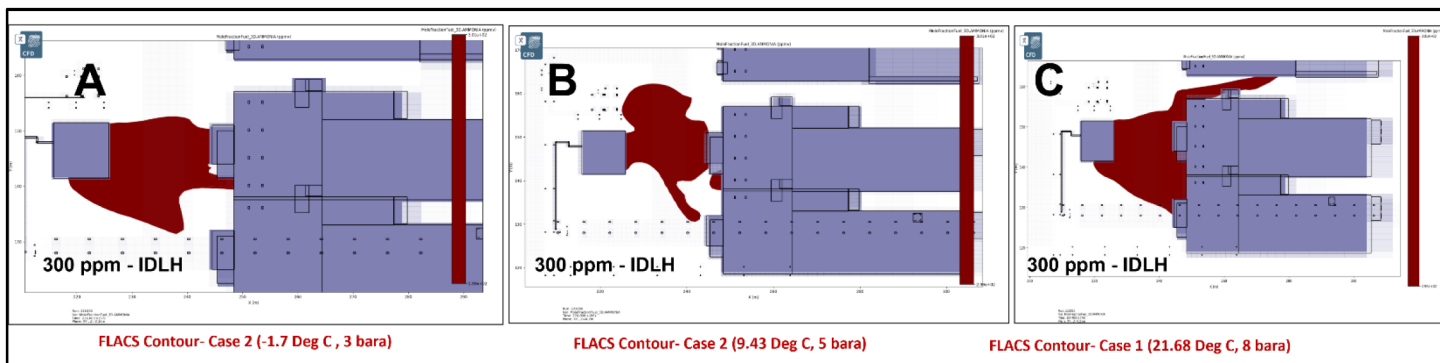


Figure 5—Ammonia 100 ppm IDLH gas dispersion counter from FLACS CFD 3D counter plot for with obstacle condition shows orthogonal (crosswind) spread increases because obstacles cause turbulence, eddy formation, and gas accumulation near surfaces.

This analysis highlights the limitation of integral/empirical tools like EFFECTS in accurately representing real-world conditions in congested industrial environments.

Case 2: Comparison of Integral/Empirical (PHAST) vs. 3D CFD (FLACS)

Similarly, PHAST's STM was used to simulate dispersion in both PHAST and FLACS. PHAST predicted the longest downwind effect distances among the tools, again due to its simplified 1D/2D approach and lack of consideration for three-dimensional site features or obstacles and can be seen in the Fig. 6 and Table 6.

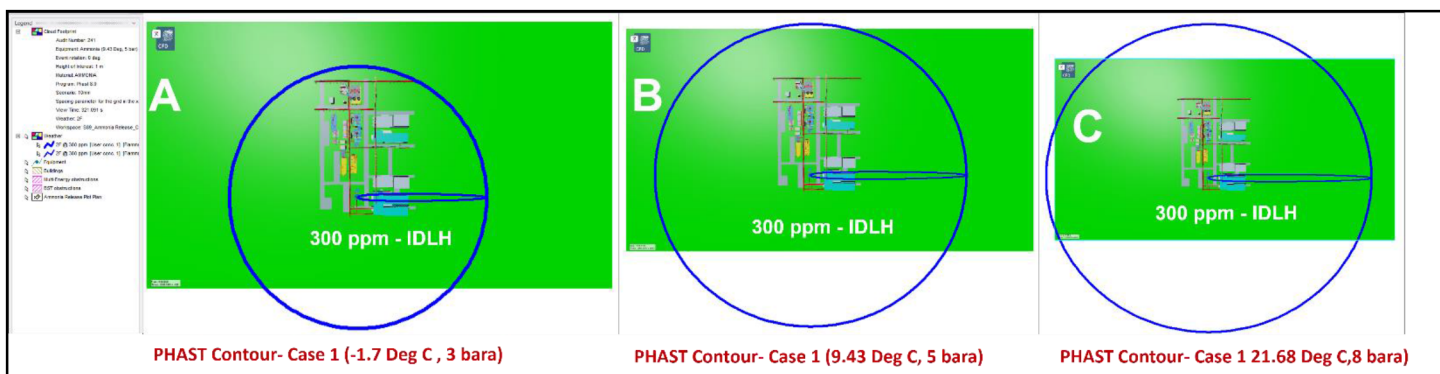


Figure 6—Ammonia 100 ppm IDLH gas dispersion counter from PHAST (Integral/Empirical) tool on Layout shows substantially longer dispersion distances because it does not consider physical obstacles that restrict gas movement

Table 6—PHAST (Integral/Empirical) Dispersion Distance Results - 2D Layout

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
1	5	-1.7	3	0.0478	190.7	10.4
2	5	9.43	5	0.07202	263.8	12.76
3	5	21.68	8	0.10777	330.1	15.25

FLACS simulations without obstacles showed substantially shorter dispersion distances, emphasizing the importance of 3D turbulence and atmospheric modeling can be seen in the Fig. 7 and Table 7

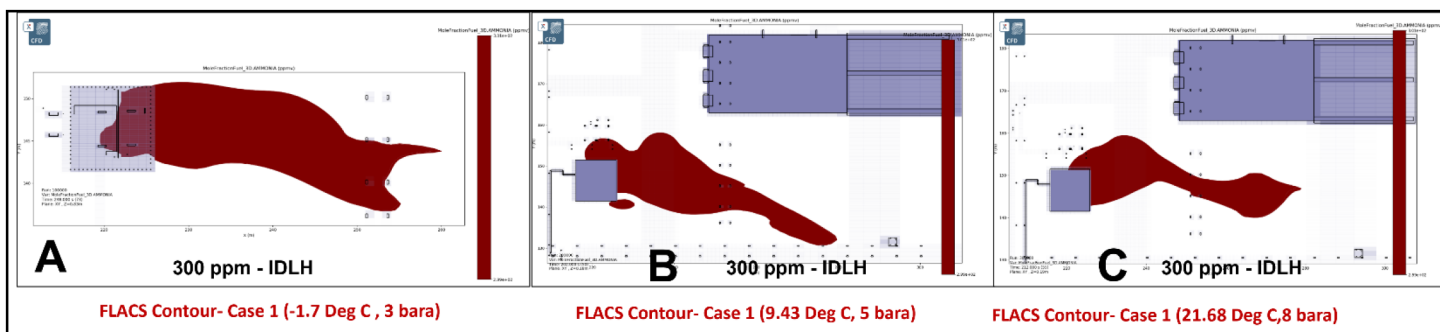


Figure 7—Ammonia 100 ppm IDLH gas dispersion counter from FLACS CFD 3D counter plot for no obstacle condition shows downwind distances are considerably shorter indicating the impact of full 3D atmospheric and turbulence modeling.

Table 7—FLACS CFD - No Obstacles Condition (Using STM from PHAST) Dispersion Distance Results

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
000111	5	-1.7	3	0.0478	41	9.5
000222	5	9.43	5	0.07202	59	25.5
000333	5	21.68	8	0.10777	61	17

Including obstacles in FLACS further reduced downwind distances but increased crosswind spread for the same reasons as observed in Case 1 and can be seen in the Fig. 8 and Table 8.

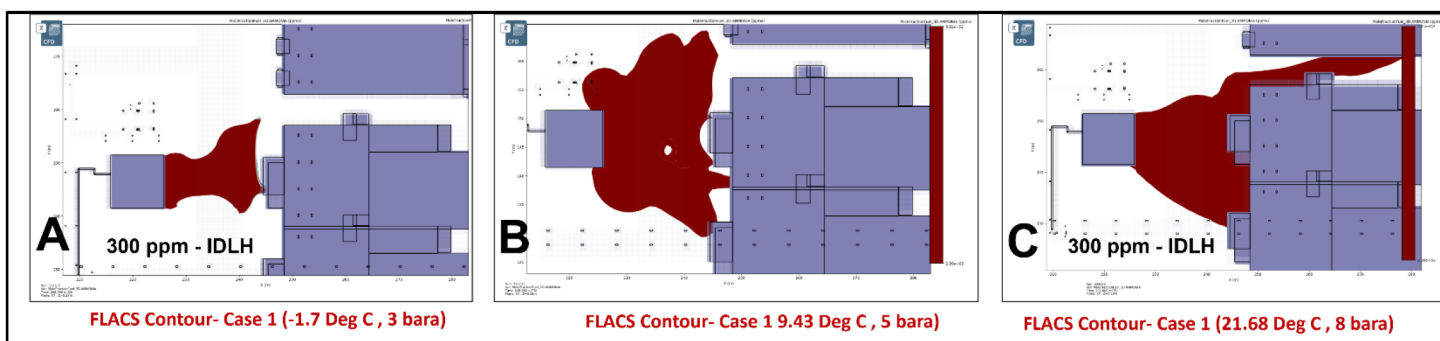


Figure 8—Ammonia 100 ppm IDLH gas dispersion counter from FLACS CFD 3D counter plot for with obstacle condition shows induced obstacles in FLACS further reduced downwind distances but increased crosswind spread for the same reasons

Table 8—FLACS CFD - With Obstacles Condition (Using STM from PHAST) Dispersion Distance Results

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
111000	5	-1.7	3	0.0478	24	18
222000	5	9.43	5	0.07202	26	30.2
333000	5	21.68	8	0.10777	58.8	32

Note: The FLACS and EFFECTS simulations were performed using Gexcon's proprietary licenses, while PHAST simulations utilized a license provided by the ADNOC offshore and UPL University research group.

Case 3: Source Term Model (STM) from 3D CFD FLACS with and without physical obstacles, terrain, and crosswind effects

In this case, dispersion simulations were conducted using 3D CFD FLACS to evaluate the influence of crosswind and terrain on toxic gas cloud behavior. These parameters cannot be adequately represented in conventional 2D empirical or integral models, which, despite their simplicity and low computational requirements, fail to capture complex three-dimensional flow interactions. Such limitations become critical in scenarios involving opposing or transverse winds, minimal crosswind diffusion, low wind speeds (<1 m/s), multiple adjacent release sources, or the presence of terrain features and obstacles such as process structures or equipment.

In the crosswind simulation, the plume was initially observed to travel parallel to the leak axis, driven by the high momentum of the release. Upon interacting with the crosswind, the plume deflected significantly, dispersing toward the upwind side and expanding coverage beyond the expected downwind footprint and can be seen in the Fig. 9 and Table 9

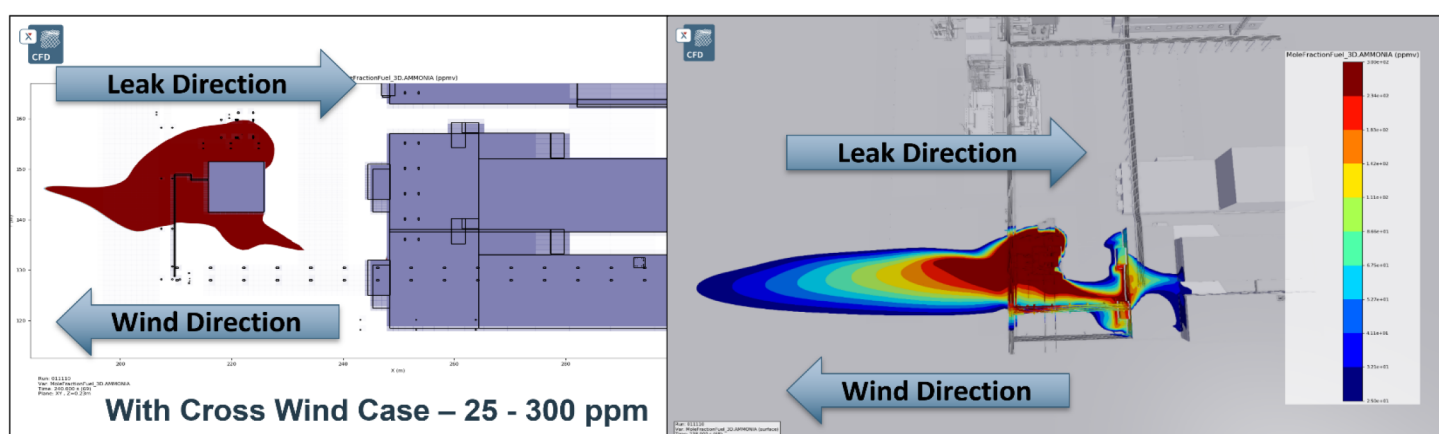


Figure 9—Ammonia 100 ppm IDLH gas dispersion counter from FLACS CFD 3D counter plot for with cross wind condition shows the plume deflected significantly, dispersing toward the upwind side and expanding coverage beyond the expected downwind footprint involving opposing or transverse winds

Table 9—FLACS CFD - Cross Wind Dispersion Distance Results

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
011110	5	-1.7	3	0.0478	41	9.5

For the terrain-influenced simulation, the downwind dispersion distance was shorter compared to the flat-ground scenarios in Case-1 and Case-2. The elevated ground caused the plume to rise and disperse upward, limiting its horizontal travel distance while increasing vertical mixing and can be seen in the Fig. 10 and Table 10.

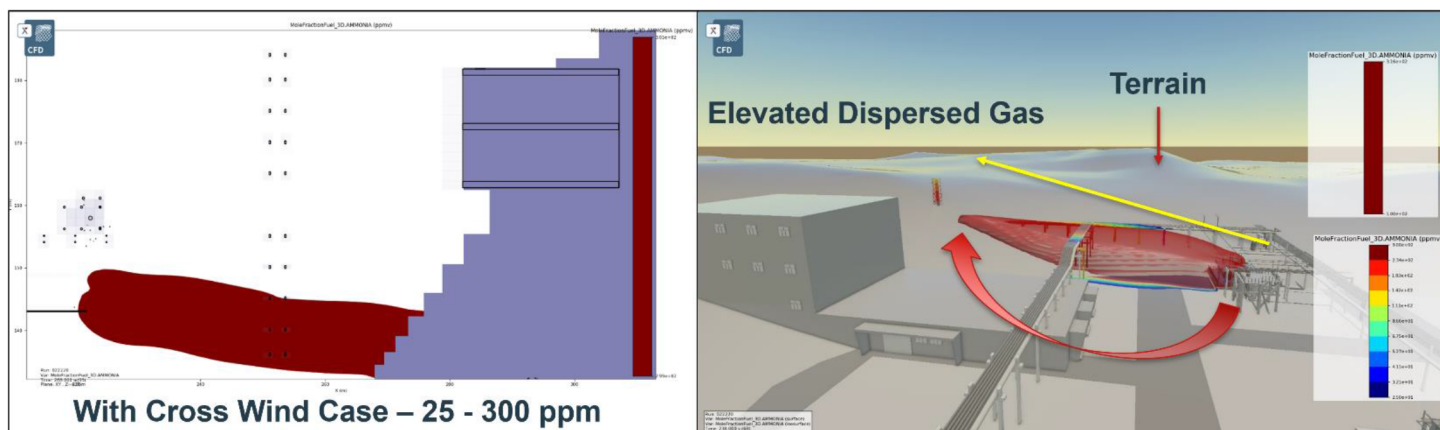


Figure 10—Ammonia 100 ppm IDLH gas dispersion counter from FLACS CFD 3D counter plot for with Terrain condition shows elevated ground caused the ammonia plume to rise and disperse upward, limiting its horizontal travel distance while increasing vertical mixing

Table 10—FLACS CFD - Gas dispersion distance result with Terrain Distance

Case	Volume (m ³)	Temp (°C)	Pressure (bara)	Release Rate	Effect Distance (m)	Width (m)
022220	5	-1.7	3	0.0478	24	18

This behavior highlights the necessity of using 3D CFD modeling to capture the combined effects of terrain elevation, obstacle interference, and wind direction variability on toxic dispersion, as supported by the quantitative results provided in the accompanying table.

Conclusion

This study has compared empirical, integral, and advanced 3D CFD approaches for modelling ammonia dispersion, revealing clear differences in predictive capability and practical applicability. While 2D tools such as EFFECTS and PHAST offer speed and simplicity for preliminary, open-space assessments, they consistently overestimate downwind dispersion distances when obstacles, terrain complexity, or crosswind conditions are present. These over-predictions stem from the simplifying assumptions in Gaussian-type formulations, which neglect the influence of congestion, low wind speeds, variable meteorological conditions, and physical flow disruptions.

In contrast, 3D CFD modelling with FLACS delivers a more realistic and physically accurate representation of plume behavior. By resolving turbulence, eddies, gas accumulation, and complex wind-obstacle interactions, CFD captures dispersion patterns that 2D tools cannot. The results show that physical obstacles reduce downwind plume travel—often to one-third of the unobstructed distance—while increasing lateral spread due to flow disruption. In crosswind scenarios, CFD simulations reveal plume reversal into upwind regions, a phenomenon entirely missed by 2D models. Terrain effects were also shown to lift and shorten the downwind travel of ammonia clouds, highlighting the limitations of assuming flat, unobstructed domains. Furthermore, the detailed summary [Table 11](#) is given below for more comprehensive understanding

Table 11—Comparative Evaluation of Ammonia Dispersion Modeling Approaches

Criteria	EFFECTS / PHAST (Empirical & Integral)	FLACS-CFD (3D CFD)
Accuracy	• Moderate in complex scenarios. • Tends to overestimate downwind dispersion distances due to simplified Gaussian assumptions. • Cannot capture turbulence, terrain, or obstacles. • Suitable only for open, flat terrain with steady winds. • Often over-conservative in predicting toxic zones.	• High fidelity. Resolves turbulence, eddies, gas accumulation, plume lifting, crosswind reversals, and mitigation measures. • Realistic prediction of hazard zones in complex industrial and urban environments. • Supported by extensive validation
Computational Time	• Low to moderate (minutes).	• Fast/Quick – Using High Computational Power (HPC) but costly • High (hours–days depending on mesh/domain size).
Applicability	• Detailed consequence analysis for safety reports • Design studies • QRA screening • Regulatory submissions	• Complex site-specific studies • terrain/obstacle effects • mitigation evaluation (water curtains, deluge)
	• Detailed design studies in unobstructed environments • Useful as an intermediate step before CFD.	• Essential for Inherently Safer Design (ISD) • ALARP demonstration • emergency planning.
Limitations	• Cannot model crosswind or complex terrain- Neglects obstacles and mitigation • Over-simplified dispersion physics. • Cannot handle transient or crosswind effects well • Limited for multi-source and low wind (<1 m/s) cases • Obstacle interaction not captured. • humidity effects. - Cannot capture plume recirculation or lateral spread	• High computational demand • Requires expertise for setup • Time Consuming
Findings from This Study	• Consistently over-predicted ammonia dispersion length in crosswind/obstacle scenarios.	• Accurately captured shorter downwind travel (up to one-third of 2D predictions), greater lateral spread, and plume reversal. Demonstrated critical terrain effects (lifting, shortening of plume).

These findings strongly align with existing literature, which has consistently reported the inability of integral and empirical models to capture realistic dispersion behavior in congested industrial environments. Prior studies by Efthimiou et al. (2017)²¹, Hansen et al. (2010)²⁸, and others confirm that CFD's ability to incorporate site geometry, terrain elevation, and dynamic wind conditions results in more accurate hazard zone predictions. This accuracy is critical in safety-critical sectors where over-conservatism can drive unnecessary capital expenditure, while under-prediction may compromise emergency preparedness and mitigation effectiveness.

From a risk management perspective, selecting the appropriate modelling approach is not merely a technical decision it directly influences inherently safer design (ISD), risk-based process safety strategies,

and the achievement of ALARP risk profiles. In the context of the energy transition, where ammonia is gaining prominence as a zero-carbon fuel, accurate consequence modelling will be central to ensuring the safe deployment of storage, handling, and transport infrastructure. The ability of CFD to reduce modelling assumptions while accurately predicting dispersion in complex real-world conditions makes it indispensable for designing effective mitigation systems, zoning hazardous areas, and optimizing emergency response plans.

In summary, the integration of this study's findings with established research reinforces the necessity of incorporating advanced 3D CFD into consequence analysis for ammonia releases in congested or terrain-affected environments. As industrial operations grow in complexity and public safety expectations rise, reliance on CFD tools like FLACS will be essential for safeguarding human health, protecting the environment, and optimizing resource allocation ensuring safety without unnecessary overdesign.

Future Work

Although ammonia is primarily recognized as a toxic gas, under certain conditions it also exhibits flammable and explosive behavior. Ammonia has a flammability range in air of approximately 15-28% by volume, with an ignition temperature around 651 °C³⁹. Its relatively high ignition energy and narrow flammability limits mean that accidental ignition is less likely compared to common hydrocarbons. However, in confined or congested environments, where ammonia can accumulate and mix within the flammable range, it can act as a fuel source for fires and explosions⁴⁰.

In particular, pressurized releases, enclosed spaces, or scenarios involving hot surfaces and ignition sources significantly increase the potential for ammonia vapor to ignite or detonate⁴¹. Additionally, under certain conditions, ammonia can undergo deflagration-to- detonation transition (DDT), resulting in hazardous overpressures capable of damaging equipment and posing risks to personnel.

For these reasons, while the current study focused on dispersion, future work must also address ammonia's flammable and explosion characteristics to provide a more comprehensive risk assessment. Integrating dispersion and explosion modeling will be essential for safer design, effective risk reduction, and informed decision-making in ammonia storage, transport, and energy transition applications⁴².

Disclaimer

The primary objective of this paper is to demonstrate the importance of selecting appropriate and accurate analysis techniques. Both integral/empirical models (EFFECTS, PHAST, etc.) and 3D CFD (FLACS) tools are technically sound, and their results can be applied as per specific study requirements. This work is not intended to compare the superiority of one software over another, nor to market or promote any particular tool. The sole purpose is to highlight correct methodologies for simplifying consequence analysis studies. All simulations presented are based on hypothetical scenarios and are provided for learning and understanding purposes only. These results must not be applied directly to any real facility for safety-related decision-making.

Credit authorship contribution statement

- **Govind K. Patil:** Conceptualization, CFD modelling (FLACS), Empirical/Integral modelling (EFFECTS), Methodology, Investigation, Writing - original draft.
- **Rajkumar Lodha:** Conceptualization, Empirical/Integral modelling (PHAST), Methodology, Investigation, Comparative analysis - CFD vs. Empirical/Integral modelling, Writing - review & editing.
- **Jitendra B. Naik:** Supervision, Conceptualization, Writing - review & editing.
- **Nikita K. Patil:** CFD post-processing, Literature review, Draft writing.

- **Omprakash K. Mahadwad:** Empirical/Integral modelling (PHAST), Writing - empirical/integral modelling section.
- **Ravindra B. Kanawade:** Empirical/Integral modelling (PHAST), Writing - empirical/integral modelling section.

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Nomenclature / List of abbreviations

Abbreviation / Symbol Full Form / Description

1D	One-Dimensional
2D	Two-Dimensional
3D	Three-Dimensional
3D-CFD	Three-Dimensional Computational Fluid Dynamics
ALOHA	Areal Locations of Hazardous Atmospheres (dispersion model)
ALARP	As Low As Reasonably Practicable — risk management principle
CAPEX	Capital Expenditure
CFD	Computational Fluid Dynamics
CFLC	Courant–Friedrichs–Lewy number based on acoustic speed
CFLV	Courant–Friedrichs–Lewy number based on flow velocity
DNV	Det Norske Veritas (classification and risk software)
EFFECT	Gexcon's EFFECT software (Gaussian-based dispersion model)
ER	Emergency Response
FLACS	FLame ACceleration Simulator (Gexcon's CFD tool)
GHG	Greenhouse Gas
HSE	Health, Safety, and Environment
IDLH	Immediately Dangerous to Life or Health
IMO	International Maritime Organization
ISD	Inherently Safer Design — design approach to reduce hazards
LEL	Lower Explosive Limit
LOC	Loss of Containment
NIOSH	National Institute for Occupational Safety and Health
NOx	Nitrogen Oxides
N ₂ O	Nitrous Oxide
PHAST	Process Hazard Analysis Software Tool
PPE	Personal Protective Equipment
ppm	Parts Per Million
QRA	Quantitative Risk Assessment
RANS	Reynolds-Averaged Navier–Stokes equations
RTI	Relative Turbulence Intensity
STM	Source Term Data
TLS	Turbulence Length Scale

TNT Eq.	Trinitrotoluene Equivalent
UEL	Upper Explosive Limit
VCE	Vapour Cloud Explosion
WIND	Boundary condition representing wind inflow and parallel boundaries
NOZZLE	Boundary condition representing outlet (exit) boundary
EULER	Eulerian numerical approach for fluid simulation
k	Turbulent kinetic energy
ε (ε)	Rate of dissipation of turbulent kinetic energy
h	Enthalpy (heat content per unit mass)
Y_F	Fuel mass fraction
ξ_{xi}	Mixture fraction (fuel-oxidizer mixing ratio)
$C_\mu, C_1\varepsilon, C_2\varepsilon, \sigma_k, \sigma_\varepsilon$	Constants in the standard $k-\varepsilon$ turbulence model

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